CHEMICAL ENGINEERING DEPARTMENT WEBSITE

HTML5 Mini Project Report

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Register Number	TRP26CS047
Section	CSE A
Allocated Topic	Department of Chemical Engineering
College	Add official college name
Technology	HTML5, CSS3, SVG, XAMPP / Apache

Prepared as an academic website development assignment. Complete the environment details, genuine screenshots and testing results before final submission.

1. Introduction

This project is a six-page website for a college Department of Chemical Engineering. It introduces the discipline, presents illustrative course areas, provides a registration/enquiry form, lists sample contact information and explains technical evidence. It uses semantic HTML5 and basic CSS without JavaScript frameworks.

2. Requirement Analysis

The website includes index, about, courses, register, contact and evidence pages. Users navigate through shared links. Courses includes a captioned table; Register uses labelled fields and native browser validation; local SVGs provide visual media with alternatives. Full analysis is in documents/requirement-analysis.md.

3. Site Map and Web Concepts

Home links to About, Courses, Register, Contact and Evidence. The Internet is a global network of interconnected networks; the Web is a service for accessing linked resources using HTTP/HTTPS. A browser requests HTML and assets from a server. DNS maps domain names to IP addresses, while localhost refers to the local computer. Apache serves the project locally.

4. URL Analysis

Example: http://localhost/chemical_engineering_department_project/courses.html. The scheme is http, host is localhost, default port is usually 80, and the path identifies the project folder and page. Relative paths such as css/style.css support portability. See documents/url-analysis.md.

5. Development Environment and Tools

Recommended environment: XAMPP with Apache. Copy the project into htdocs, start Apache and open the localhost URL. Use Elements to inspect HTML, Network to inspect HTML/CSS/SVG requests and status codes, and Console to check browser messages. Insert genuine screenshots into the supplied folders.

6. Accessibility, Table, Form and Media

Semantic landmarks, skip link and keyboard focus styling support accessibility. The table has a caption and row/column headers. Form labels match control IDs, and required fields, email, phone and date inputs demonstrate native validation. Local SVG illustrations include descriptions. The form does not save or send information.

7. Hosting and Security

A public GitHub repository is suitable for source review. Use faculty-approved static hosting if public deployment is required. Do not expose credentials or private data. Browser validation is not a security boundary; production systems require server-side validation, HTTPS, access controls and safe data handling. See documents/security-and-hosting.md.

8. Testing Record

T01–T12 cover localhost access, navigation, CSS/SVG loading, form validation, responsive layout and Developer Tools. Actual-result and status fields remain unfilled until testing is performed. Record defects and retest after corrections.

9. Reflection (200–300 words)

Building the Department of Chemical Engineering website helped me understand how a multi-page website is planned and structured with HTML5. I divided the content into six pages so that each page has a clear purpose: home, about, courses, registration, contact and evidence. A shared navigation menu and footer make it easier for users to move between pages and keep the design consistent.

I used semantic elements such as `header`, `nav`, `main`, `section`, `article`, `table`, `form` and `footer`. These elements make the document structure clearer than using generic containers everywhere. The course table includes a caption and appropriate row and column headers. The enquiry form uses labels connected to controls, suitable input types and built-in browser validation. I also included a skip link and visible keyboard focus styling to improve accessibility.

The project uses relative paths for pages, CSS and SVG illustrations. This makes it easier to move the project folder to a local server such as XAMPP. Developer Tools can be used to inspect the document structure, verify resource requests and identify console errors. The testing record provides a systematic way to check navigation, images, form constraints and responsive layout.

One limitation is that the form is only a front-end demonstration. It does not store or send information because no backend has been included. The department vision, contact details and course information are also sample content and must be checked against official college information. Overall, this project gave me practical experience in planning, writing, organizing and testing a small website. It also showed me why accessibility, accurate content, security awareness and honest evidence are important parts of web development.

10. Known Limitations

The form has no backend and does not persist or transmit data. Department vision, curriculum, college name and contact details are illustrative placeholders and must be verified. No login, database, admin dashboard or dynamic updates are implemented.

11. References

WHATWG HTML Living Standard: https://html.spec.whatwg.org/ ; MDN HTML: https://developer.mozilla.org/en-US/docs/Web/HTML ; MDN Forms: https://developer.mozilla.org/en-US/docs/Learn/Forms ; MDN CSS: https://developer.mozilla.org/en-US/docs/Web/CSS ; Apache Friends XAMPP: https://www.apachefriends.org/ ; W3C WAI Forms: https://www.w3.org/WAI/tutorials/forms/.

Add access dates after consulting each source.

Submission checklist

■ Replace college and placeholder information.

■ Run all six pages through localhost.

■ Capture genuine screenshots for localhost, Elements, Network and Console.

■ Complete T01–T12 actual results and statuses.

■ Review the report and upload the project to GitHub.